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916 Ma Pole for southwestern Baltica: palaeomagnetism of the Bjerkreim-Sokndal layered intrusion, Rogaland Igneous Complex, southern Norway

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SUMMARY

The Rogaland Igneous Complex (RIC) in southern Norway intruded into post-Sveconorwegian granulite facies crust ∼930 Ma. It includes three massif anorthosites, several small leuconorite bodies and the ∼7 km thick norite-quartz mangerite layered Bjerkreim-Sokndal (BKS) intrusion. The intrusion consists of five rhythmic megaunits created by repeated magma influxes topped by a transition zone and more evolved mangerites and quartz mangerites. Over 70 palaeomagnetic sites have been collected in the BKS, sampling all the megacyclic subunits and overlying mangerites. Remanence within the BKS is held in hemo-ilmenite-only rocks (lower parts of the megacyclic units), mixed hemo-ilmenite and magnetite rocks (upper parts of the lower megacyclic units) and magnetite only rocks in the upper highest megacyclic unit and overlying mangerites. Due to the different oxides present magnetic susceptibility varies over four orders of magnitude with a bimodal distribution (mean susceptibility of 6.4×10^{-3} SI for hemo-ilmenite rocks, and 8.7×10^{-2} SI for magnetite rocks). NRM values do not show a strong bimodal distribution as many of the rocks lacking magnetite have hemo-ilmenite with strong lamellar magnetism; average NRM for the entire suite is 8.83 A m^{-1}. All sites within the cyclic part of the intrusion have stable remanence and produce well-clustered site means. Samples from the upper mangerite rocks, dominated by MD magnetite, are commonly unstable and not all sites provide acceptable data. Mean directions for 66 sites spanning the entire intrusion are $I = -73.5^\circ$, $D = 303.4^\circ$, with $\alpha_{95} = 3.7^\circ$ and $k = 24$. The resulting pole position is at 35.9°S and 217.9°E, with a palaeolatitude for this part of Baltica of −59.4°. Examination of the magnetic mineralogy combined with geochronology for RIC rocks and cooling rates for the region yields an age of magnetization of 916 Ma. Metamorphic country rocks yield similar directions at least 10 km from the contact, confirming the presence of a contact aureole around the intrusion. Comparison to other early Neoproterozoic palaeomagnetic poles from southern Scandinavia confirms the high southern latitude position of Baltica at this time, and combined with the few other ∼900 Ma poles from Baltica affirm the presence of the Rodinia supercontinent at this time.

Key words: Magnetic mineralogy and petrology; Palaeomagnetism applied to tectonics; Rock and mineral magnetism.

1 INTRODUCTION

Rodinia, the supercontinent, named by McMenamin & McMenamin (1990), existed for some 500 million years from the late Mesoproterozoic through the Neoproterozoic, although exact timing of amalgamation and breakup are under continued discussion (e.g. Evans 2013). During this time Rodinia contained all the major continental blocks, although the combinations and juxtapositions of individual blocks remains enigmatic (Li *et al.* 2008; Evans 2009). The classic distribution of cratons includes Laurentia, Baltica and Amazonia juxtaposed by the prominent ∼1.0 byr orogenic event, the Grenville Orogeny in Laurentia, the Sveconorwegian in Baltica and the Sunsas in Amazonia (Hoffman 1991; Johansson 2009). The position of the participating cratons is commonly controlled and/or constrained by palaeomagnetic data from suitable dated rocks (e.g. Meert & Torsvik 2003; Li *et al.* 2008). The selection process for usable geomagnetic pole positions is subjective, although some authors consider the quality factor established by Van der Voo (1990)

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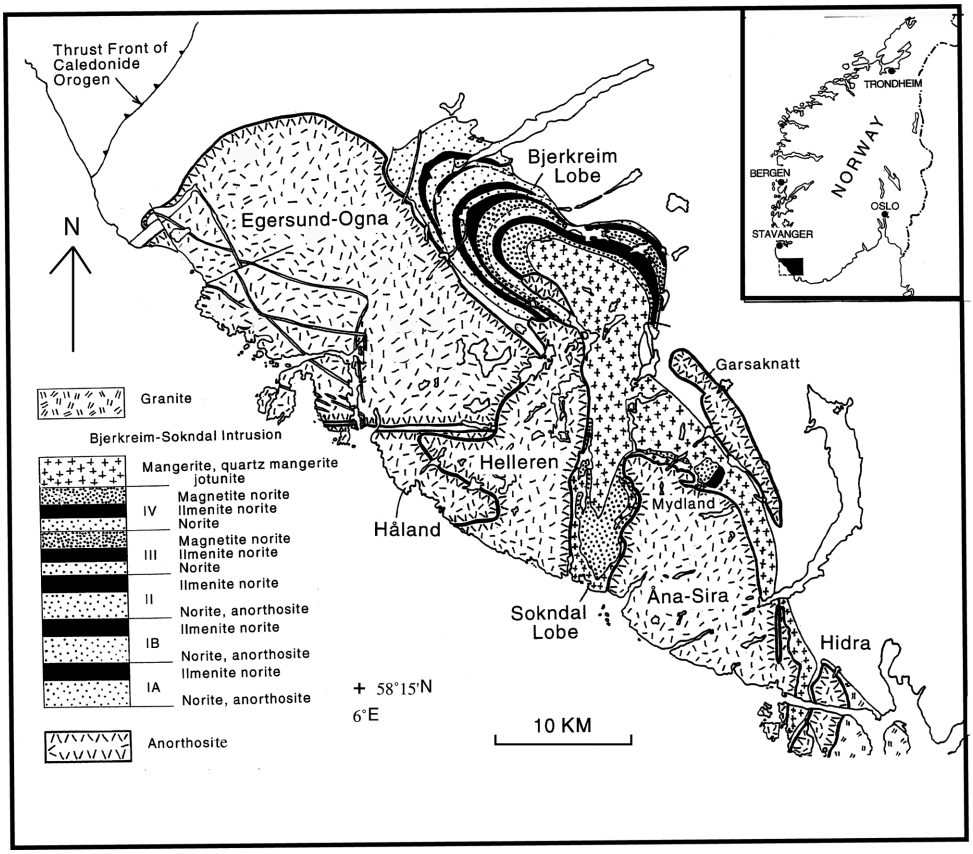


Figure 1. Generalized geologic map of the Rogaland Igneous Complex, southern Norway. Adapted from Brown & McEnroe (2004).

or follow stricter specific criteria for 'key poles' (Buchan *et al.* 2000; Buchan 2013). Depending on the criteria used, poles numbers for the entirety of Rodinia can vary from over 100 (Li *et al.* 2008) to less than 20 (Buchan 2013).

The position of Baltica during the formation and breakup of Rodinia remains difficult to assess, due to the lack of well-dated reliable palaeomagnetic results from Baltica during the Neoproterozoic. Although the position of Baltica in the Early Ediacaran (~600 Ma) is better constrained (Meert 2014), the earlier part of the Neoproterozoic is not well represented. Even the gross position of Baltica is questioned—was it in its current position, relative to Laurentia ('right-side up') or had the entire crustal block been rotated to be 'upside-down' (Hartz & Torsvik 2002; Cawood & Pisarevsky 2006)? Possible lesser rotations have also been suggested in the positioning of Baltica relative to Laurentia (Torsvik *et al.* 1996; Bogdanova *et al.* 2008). More recently it has been suggested that the southwestern margin of Baltica may not represent a continent/continent collision, as assumed in earlier configurations but rather it is an accretionary margin (Slagstad *et al.* 2013), thus allowing for configurations without another craton outboard.

To increase the information available for plate reconstructions, orientations and apparent polar wander paths, we need to provide well-dated, primary directions from established cratons, in this case Baltica. In particular, when dealing with Proterozoic rocks, a sufficient numbers of samples and sites are necessary to provide statistically suitable data. Here, we investigate the palaeomagnetism of rocks from the Rogaland Igneous Complex (RIC) in southernmost Norway, where the 7 km thick Bjerkreim-Sokndal (BKS) layered intrusion was emplaced into the post-Sveconorwegian terrain of southern Norway at 931 Ma. Palaeomagnetic results from this well-studied layered intrusion add important information and data for future models of the position of Baltica within the supercontinent Rodinia.

2 GEOLOGIC SETTING

The Neoproterozoic RIC intrudes granulite facies metamorphic rocks within the Baltic shield in southern Norway (Fig. 1). The RIC has an estimated volume of some 5000 km^3 (Schärer *et al.* 1996), although modelling of offshore gravity and magnetic data indicate the largest part of the complex extends south into the Norwegian–Danish Basin (Olesen *et al.* 2004). The surrounding metamorphosed country rock, dated at ~980 Ma (Bingen & Van Breemen 1998; Bingen & Stein 2003), represents the end of the Sveconorwegian (Grenville) orogeny, 1.03–0.97 Ga, evident in much of southern Baltica. The RIC consists of three large 'massif' anorthosite bodies (Egersund-Ogna, Håland-Helleren and Åna Sira), the BKS layered intrusion and several smaller intrusive bodies including the Farsund charnockite and the Garsaknatt leuconorite. U-Pb zircon ages from the three large anorthosites are tightly clustered at 931 ± 2 Ma (Schärer *et al.* 1996) and samples from the fayalite quartz mangerite of the BKS yield an emplacement age 931 ± 7 Ma, indicating intrusion some 60 million years after the last deformation in the area (Vander Auwera *et al.* 2011). Additional ages on norite dikes cross-cutting the intrusions, and one distinct quartz mangerite at the top of the layered intrusion yield younger ages close to 920 Ma (Schärer *et al.* 1996; Vander Auwera *et al.* 2011).

BKS norite-quartz mangerite layered intrusion (Duchesne 1987; Wilson *et al.* 1996; Robins & Wilson 2001) intrudes both the

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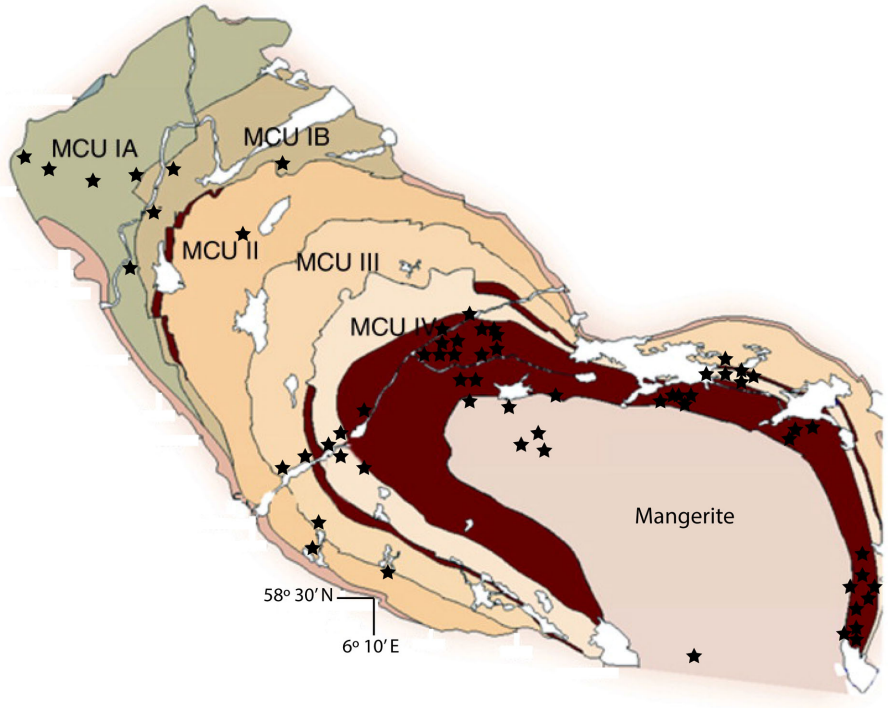


Figure 2. Location of 60 palaeomagnetic sites (stars) in the Bjerkreim lobe of the Bjerkreim-Sokndal intrusion. Additional six sites are located south of area in the Sokndal Lobe. The five cyclic units, labelled MCU IA, IB, II, III and IV, are shown, along with the transition zone, mangerite and quartz mangerite in the centre of the intrusion. Red zones are where magnetite is the primary iron oxide phase present. Map adapted from McEnroe *et al.* (2009).

granulite-facies metamorphic rocks to the north and the anorthosite bodies of the RIC to the south (Fig. 1). It has long been recognized as one of the classic 'massif' layered intrusions found in the Proterozoic (Cawthorn 1996). The areal extent of the BKS is estimated to be 230 km^2 and a thickness of ∼7 km (Wilson *et al.* 1996). The intrusion is divided into the well-exposed northern Bjerkreim lobe, and the southern Midland and Sokndal lobes (Fig. 1).

The layered series of the BKS intrusion were first described by Michot (1960) as five rhythmic units created by crystallization processes during repeated magma influxes (Fig. 2). Further refinement of the intrusion has resulted in six megacyclic units (MCUs) from the base to the top, numbered: 0, IA, IB, II, III and IV (Wilson *et al.* 1996); however, the lowest-most unit, 0, is exposed in only one small outcrop with difficult access. The five major MCUs are further divided into subunits 'a–f' based on assemblages of cumulus minerals (Robins & Wilson 2001). From the point of view of palaeomagnetic studies the crystallization sequence of the magnetic minerals, here magnetite, titanomagnetite and hemo-ilmenite, within different MCUs, and subunits, is important and is illustrated on the right-hand side of the stratigraphic column (Fig. 3). Above the last megacyclic unit (MCU IV) is a transition zone and then massive, more evolved mangerite and quartz mangerite layers. A detailed description of the megacyclic mineralogy is given by Wilson *et al.* (1996) with discussion of the magnetic oxides by McEnroe (1997) and McEnroe *et al.* (2001b, 2004a, 2009). The entire body is distinctive on aeromagnetic maps as the lower portions have strong remanent-controlled negative anomalies while the upper mangerites have positive induced anomalies (McEnroe *et al.* 2009). Within subunit MCU IVe is an additional subdivision into unit IVe′, distinguished on high remanence (>5 A m^{-1}) and large negative ground and aero-magnetic anomalies (McEnroe *et al.* 2004a,b, 2009).

The BKS is folded into a large doubly plunging syncline, with the Bjerkreim lobe occupying the southeast-plunging northern end and the Sokndal lobe occupying the north-plunging southern end. A study of the magnetic fabric of the intrusion, using the anisotropy of magnetic susceptibility, indicates that the folding probably occurred as gravity-induced subsidence related to the emplacement of neighbouring anorthosites, and occurred when the material was still in a magmatic state (Bolle *et al.* 2000). A seismic reflection profile (Deemer & Hurich 1997) and gravity modelling (Smithson & Ramberg 1979) indicate that the base of the intrusion lies at a depth of 4–5 km, but due to the early deformation, more than 7000 m of the layered intrusion has been observed (Wilson *et al.* 1996).

3 SILICATE AND MAGNETIC MINERALOGY

As discussed above, the mineralogy and petrology of the BKS layered intrusion has been studied for over 50 yr, and an excellent overview is given by Wilson *et al.* (1996). Here we focus primarily on the oxide mineralogy. However, the silicate mineralogy was examined also from all 66 localities to determine correctly which MCU, and subzone, should be assigned to each site and for the presence of oxide inclusions. The silicate mineralogy also provides an excellent template for characterizing the degree of alteration. One of the remarkable aspects of the BKS intrusion is the limited amount of alteration of the silicate or oxide mineralogy, to the extent that even the olivine in is generally in pristine condition. Fig. 4 shows photomicrographs from three zones in MCU II, III and IV. These were chosen to show variations in the silicate mineralogy in the BKS. The oldest unit shown is from MCU IIc (Fig. 4a). It resembles MCU I which is a plagioclase-rich norite, with a small amount of hemo-ilmenite. As one proceeds up the stratigraphy to

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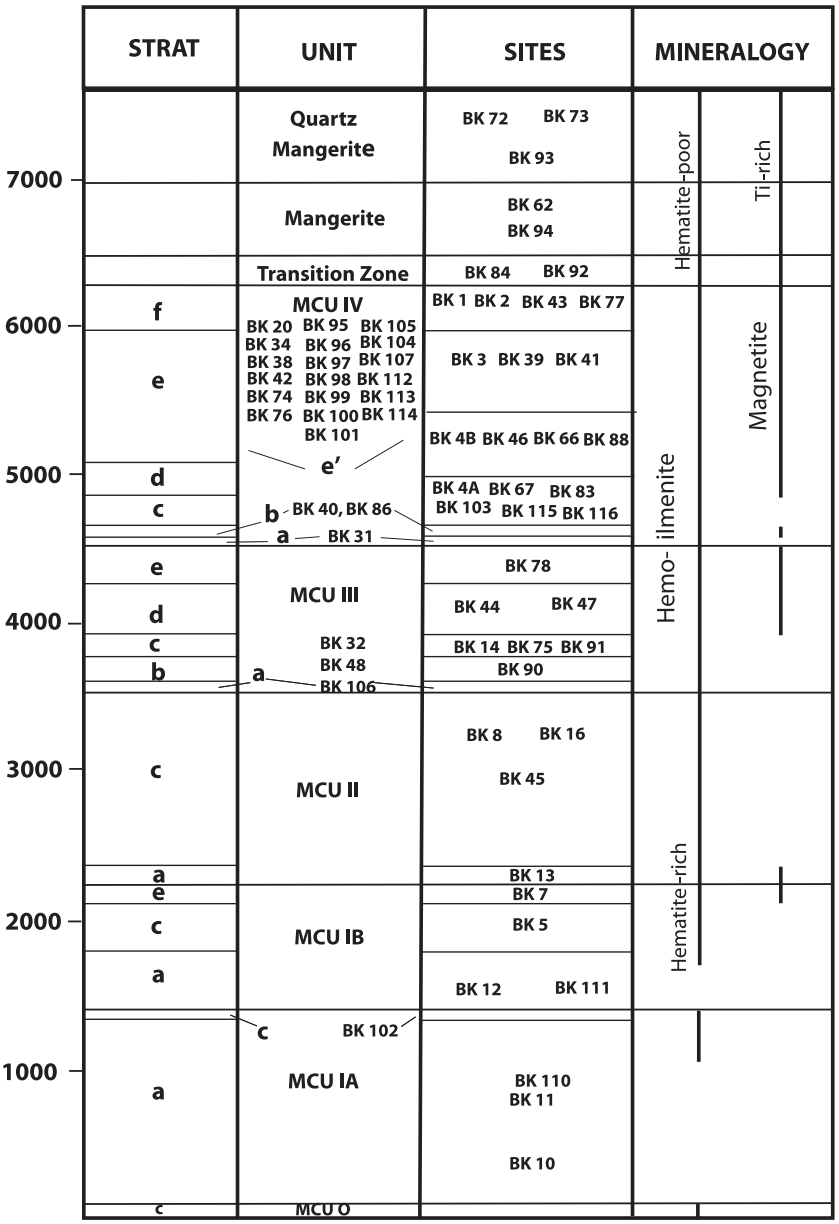


Figure 3. Stratigraphic column of the Bjerkreim-Sokndal Layered Intrusion, with megacyclic units (MCUs) and subzones (a–f). BK numbers refer to samples sites, as listed in Table 1 and shown in Fig. 2, and placed within their stratigraphic MCU position. The 19 MCU IVe′ sites are shown under the unit column but exist within the IVe subunit. Occurrence of iron oxide phases within the intrusion indicated on the left. Numbers on right are stratigraphic thickness, in metres. Adapted from Wilson *et al.* (1996) and McEnroe *et al.* (2009).

MCU II, orthopyroxene crystallizes and is common in most units, except in the recharge zone a, and subzone b in units III, IV. As shown here, the orthopyroxene is typically in excellent condition and rarely shows any sign of alteration. The orthopyroxene commonly contains ilmenite exsolution lamellae (Fig. 5a) parallel to pyroxene (100). The ilmenite exsolution contains very fine lamellae of hematite (Fig. 5b).

Fig. 4(b) is from MCU IIIb. Subzone b represents the most primitive magma associated with the intrusion (Wilson *et al.* 1996), and is the only magma to contain olivine as a primary precipitate in the MCU sequence. At the top of the intrusion, above the MCU units there are the transition zone and mangerites, and here fayalitic olivine is abundant. In MCU IIIb plagioclase, olivine, hemo-ilmenite and magnetite are the major minerals. The magnetite is commonly chrome-rich. The olivine, as shown in Fig. 4(b), is pristine, a very important observation because olivine is a mineral notorious for alteration even in an environment with minimal invading aqueous fluid.

Fig. 4(c) is from MCU IV, subzone e′. Here the major minerals are plagioclase, orthopyroxene, clinopyroxene, hemo-ilmenite, magnetite, apatite and biotite. Minor amounts of sulphides are also present. The plagioclase typically has inclusions of hemo-ilmenite as shown in Fig. 5(c). The orthopyroxene contains abundant exsolution of ilmenite (Fig. 5a) that commonly has hematite exsolution (Figs 5a and b). The clinopyroxene has exsolutions both of magnetite and hemo-ilmenite, however these magnetite lamellae are commonly of multi-domain size (Frandsen *et al.* 2004).

Earlier work on the oxides in the BKS intrusion showed that the more primitive parts of the intrusion contain hemo-ilmenite, and the more evolved magmas contained magnetite with

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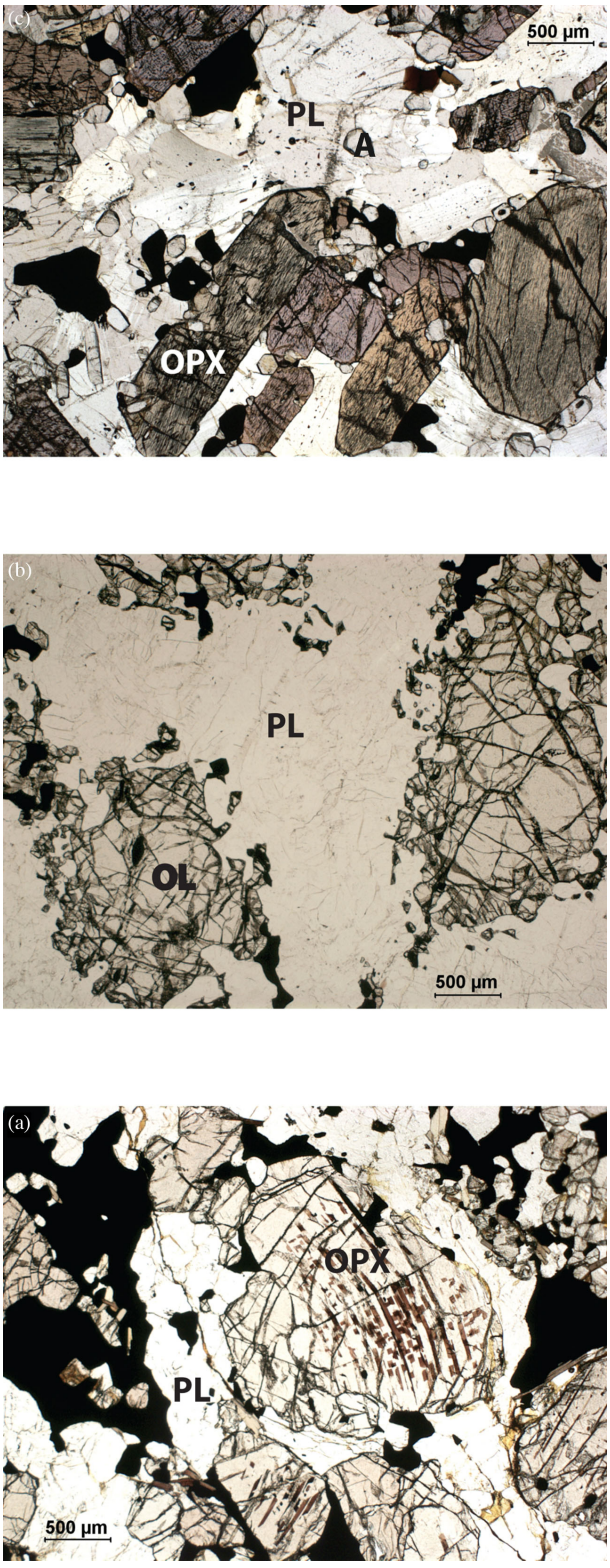


Figure 4. Transmitted-light images: (c) MCU IVe′: orthopyroxene (opx) with abundant lamellae of hemo-ilmenite, plagioclase (PL), apatite (A) opaque grains are magnetite and hemo-ilmenite. (b) MCU IIIb PL and olivine (OL), minor opaque minerals of magnetite and hemo-ilmenite. (a) Orthopyroxene with hemo-ilmenite lamellae, PL and hemo-ilmenite (opaque) from MCU IIc. See Fig. 6 for oxide mineralogy. Note 'c' is at top of image. Scale bars of 500 μm shown for each image.

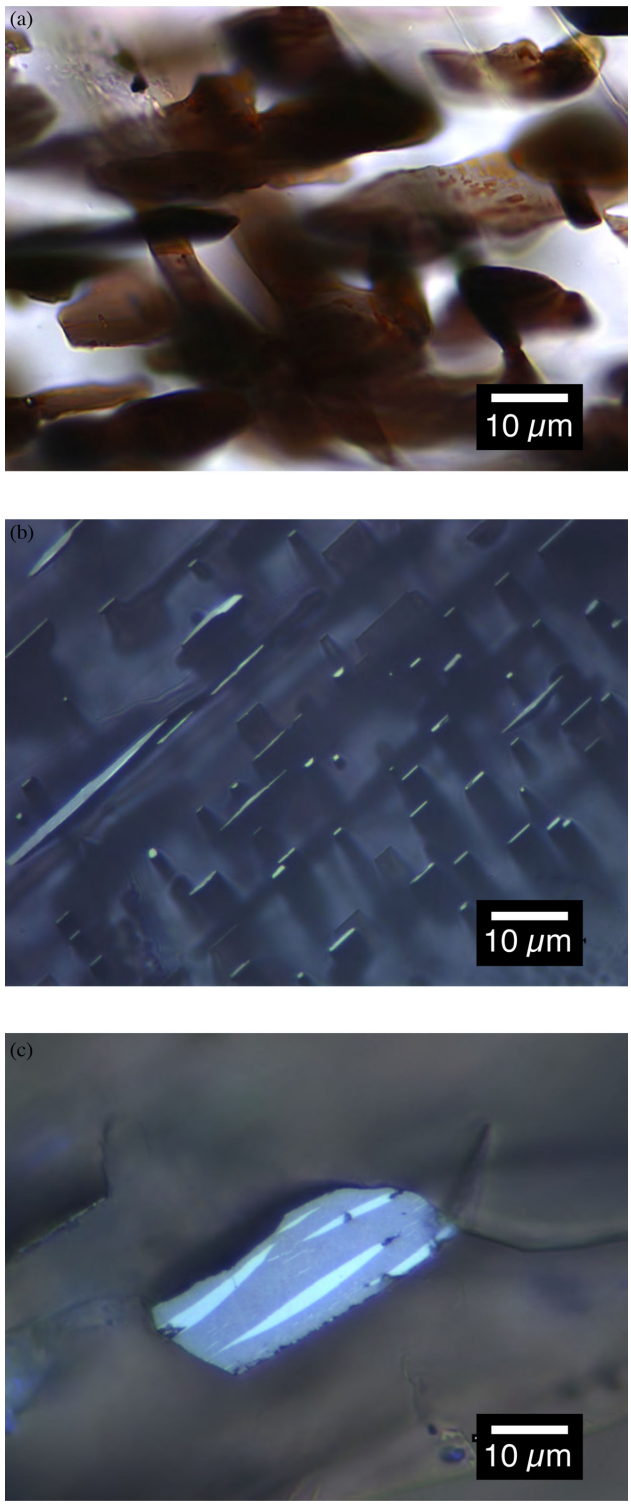


Figure 5. Reflected- and transmitted-light images of oxide inclusions in silicates. (a) Orthopyroxene with very thin lamellae of hemo-ilmenite, from MCU IVe′. Transmitted-light. (b) Reflected-light image of hemo-ilmenite lamellae in orthopyroxene. (c) Reflected-light image of discrete hemo-ilmenite inclusion in plagioclase. These inclusions are common in the lower part of the intrusion. Hemo-ilmenite lamellae commonly contain multiple generations of hematite lamellae. Scale bars of 10 μm shown for each image.

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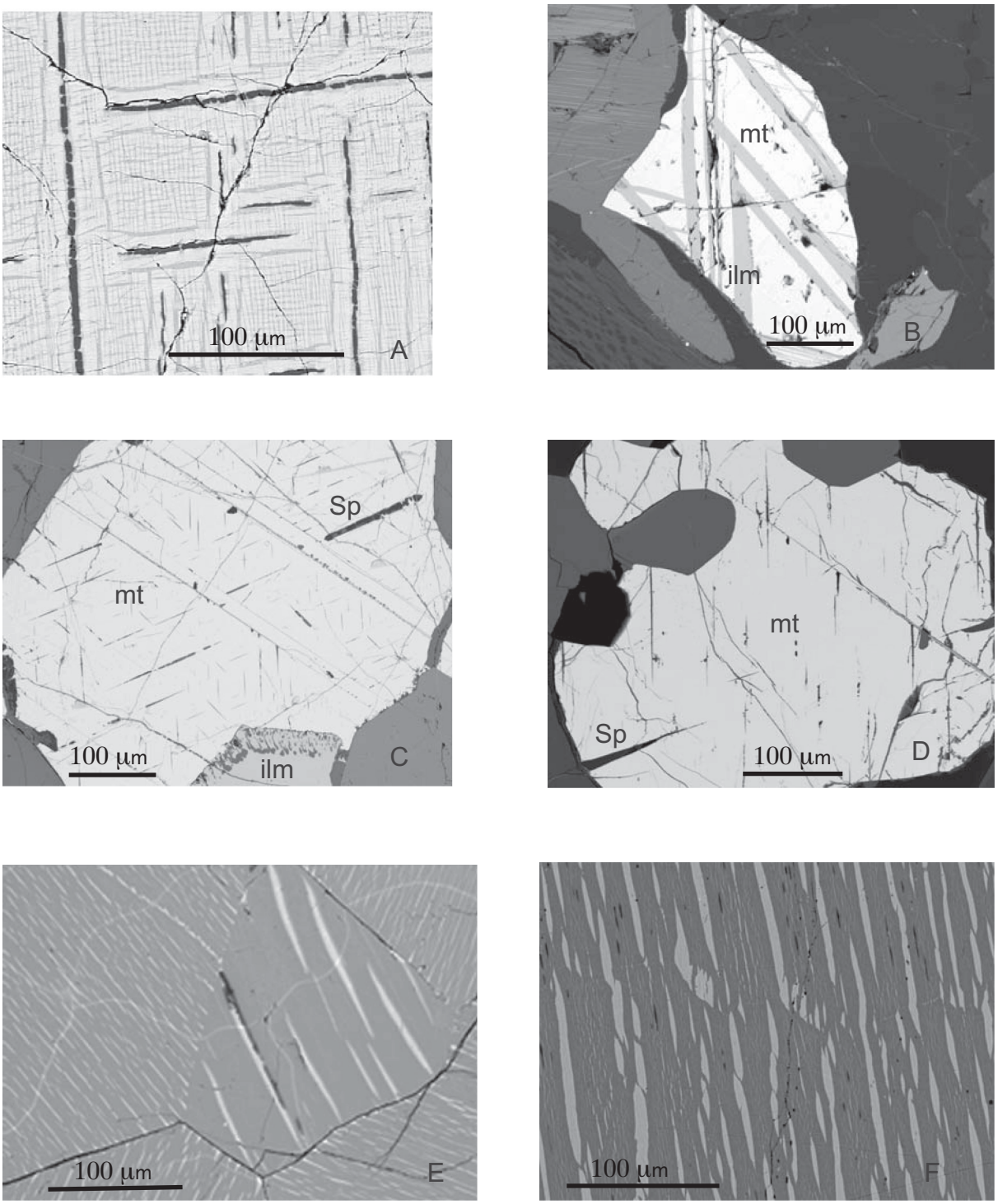


Figure 6. Electron backscatter images: (a) magnetite with 100 exsolution of ulvöspinel, later oxidized to ilmenite (light grey lamellae) and spinel needles (black) from the transition zone. (b) Magnetite with coarse sandwich type lamellae of ilmenite from mangerite. (c) Magnetite with fine ilmenite lamellae (light grey), spinel lamellae (black) and a spinel symplectite at border with large ilmenite grain, from MCU IVd. (d). Magnetite with minor oxy-exsolution of ilmenite and spinel, from MCU IVe′. (e) Hemo-ilmenite with hematite exsolution (white) from MCU IIIc, lamellae are usually <2 um wide. (f) Hemo-ilmenite with hematite exsolution (white) from MCU II. Hemo-ilmenite grains contain multiple generations of exsolution.

oxidation-exsolution of ilmenite lamellae, coexisting with near end-member ilmenite, whereas in the transition zone the magnetite is more reduced with exsolved (100) lamellae of ulvöspinel (Duchesne 1970, 1972; McEnroe *et al.* 1996, 2000, 2001b, 2009; Robinson *et al.* 2001). The ulvöspinel lamellae later oxidized to ilmenite but retain the (100) structural orientation (Fig. 6a). There is thus an evolution, from more oxidized hemo-ilmenite, with abundant hematite lamellae, to reduced magnetite, as shown by the magnetite with ulvöspinel lamellae combined with the ilmenite without hematite lamellae. This relationship with respect to the mega cyclic units is shown in Fig. 3. Duchesne (1972) and Robinson *et al.* (2001) showed that the $Fe3^{+}$ content of the ilmenite is higher in the more

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primitive parts of the intrusion, and that the ilmenite is more homogenous, with much less $Fe3^{+}$ in the evolved MCU IVf norite, and the overlying mangerite and quartz mangerite units.

Fig. 6 shows backscatter images of oxides from the different MCU layers. The most primitive layers all contain hemo-ilmenite, which commonly has multiple generations of hematite lamellae in the ilmenite (Figs 6e and f). These lamellae are rarely larger than a micron and typically continue down to the nanometre scale. Hemo-ilmenite is common throughout the intrusion in MCU I, II, III and parts of IV, though it is most abundant in subzones a and c. In the early part of the intrusion magnetite is present in MCUIB, subzone e, and in MCU II subzone a, where it can be up to 2 per cent of the rock. Except for these subzones in MCU I and II, magnetite is only present in rare amounts typically $\ll$0.1 per cent. Starting with MCU IIId magnetite was a primary precipitate and is an abundant phase in all layers above MCU IIId, except in the zone of magma mixing caused by the recharge of new more primitive magma in MCU IVa and IVc. Once magnetite appears as a primary precipitate in MCU IIId it has modal amounts ranging from 1 to 4.5 per cent. This magnetite consistently has oxidation exsolution lamella of ilmenite (Figs 6b–d), except in the transition zone where magnetite first exsolved ulvöspinel lamellae, later oxidized to ilmenite.

The rocks in BKS intrusion are coarse grained, and typically the discrete oxides are >100 um. Except for samples from the transition zone most of the oxidation-exsolution lamellae in magnetite is a trellis, or sandwich morphology, and hence the magnetite is likely to behave as multi-domain grains. The hemo-ilmenite grains, which can be >100 μm in diameter, commonly contain two, or more generations of hematite lamellae parallel to (001). Here in the BKS intrusion, the first-generation lamellae are typically thinner than 3 μm, and later generations of lamellae have submicron thicknesses. Numerous studies have shown that, with prolonged cooling, exsolution lamellae form to nanometre thickness, and even to the unit-cell thickness. Exsolution lamellae in the hematite-ilmenite solid solution series provide highly stable magnetic remanence commonly requiring fields >100 mT to demagnetize (McEnroe *et al.* 2001a, 2002, 2007a,b).

4 PALAEOMAGNETISM

4.1 Sampling

As part of a larger study of magnetic mineralogy, remanent magnetic anomalies, and magnetic behaviour in the RIC (Brown & McEnroe 2004, 2008; McEnroe *et al.* 2001b, 2004a,b, 2009), palaeomagnetic samples were collected from a large number of sites within the BKS (Fig. 2). Locations of sites and determination of the position within the MCUs (Fig. 3) were done using the geologic map of Rogaland (Marker *et al.* 2003a,b) as well as guidebooks for the region (Duchesne 2001). At many sites (∼60 per cent) samples were drilled in the field using a gasoline-powered rock drill and oriented with magnetic and sun compasses. Additional sites were collected as oriented blocks using a magnetic compass but employing both far and back sighting to obtain true orientations, with samples drilled in the lab. In all cases three to eight cores, commonly with subspecimens, were obtained. Care was taken to represent each MCU and all subunits with at least one site (Fig. 3), except for MCU 0 which is exposed only in a very limited area. Additionally, samples were collected from six sites representing the overlying transition zone, quartz mangerite, and mangerite portions of the intrusion, for a total of 66 sites for magnetic determinations. Nine sites in the polymetamorphic country rock were collected at distances between 20 and 2 km of the BKS boundary.

4.2 Laboratory procedures

Palaeomagnetic measurements were made either on a 2G Cryogenic magnetometer at the University of Massachusetts or a JR6 at the Norwegian Geological Survey (NGU). Demagnetization studies using both alternating field (AF) and thermal techniques were applied to all cores, using incremental steps. AF steps were carried out to 100 mT and thermal demagnetization went to 675 °C. Both a Molspin AF demagnetizer and ASC thermal demagnetizer were used at the University of Massachusetts, with additional measurements made at NGU using a 2G alternating field demagnetizer and a MMTD1 furnace. Susceptibility measurements were made on all cores using a Sapphire susceptibility bridge. Hysteresis parameters from at least one sample per site were measured on a Princeton Measurement Corporation 'MicroMag' Vibrating Sample Magnetometer (VSM) either at the Norwegian Geological Survey or at the Institute of Rock Magnetism at the University of Minnesota where maximum fields of 2 T could be applied.

4.3 Rock magnetic and physical properties

Detailed studies of rock magnetic and physical properties of these rocks, including susceptibility, natural remanent magnetization (NRM), Koenigsberger ratio (Q), and hysteresis loops and parameters, are described in detail by McEnroe *et al.* (2009) and will only be summarized here. Site averaged susceptibility and NRM values for all sites used in the palaeomagnetic study are included in Table 1. Overall susceptibility, determined from 66 sites, is 0.0575 SI volume, but is markedly bimodal. The primitive recharge zones, dominated by hemo-ilmenite, have an average of 0.0056 SI. Samples from the more evolved zones and the mangerites overlying the cumulate layers all contain magnetite, and have a strikingly higher mean susceptibility of 0.0872 SI. NRM values show a wide range of values, with the highest one from MCUIV-e′ where the hemo-ilmenite and magnetite samples possess the strongest remanence found in the intrusion, averaging 26.0 A m^{-1}. Average NRM for the entire BKS suite is 8.93 A m^{-1}, with a range from a minimum 0.15 A m^{-1} to a high of 60 A m^{-1}. The highest magnetized samples are from MCU IVe′, a zone with distinct magnetic mineralogy and behaviour (McEnroe *et al.* 2004a,b). Some of the lowest values of NRM are found at the top of the sequence in the quartz mangerites; these rocks have average susceptibility values (0.0248), and are dominated by MD (multi-domain) magnetite, and do not always provide stable magnetization. Q values (ratio of remanent magnetization to induced magnetization) for the entire sequence vary from 0.02 to 25, but the majority of samples have Q values greater than 0.5 (Fig. 7), indicating that remanence contributes to the resulting magnetic anomalies, as discussed by McEnroe *et al.* (2009). The three differing magnetic mineralogy combinations are evident in the Ji–Jr plot (Fig. 7) with the hemo-ilmenite samples having low induced magnetizations but high remanence, resulting in all Q values >1.0. Magnetite only sites have induced magnetizations averaging 2 A m^{-1}, but relatively low remanences, so most samples lie below the $Q = 1.0$ line. Mixed samples have induced magnetizations influenced by magnetite (average 2 A m^{-1}) but remanent magnetizations of 1–60 A m^{-1}, resulting in most samples lying above the $Q = 1$ line.

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Table 1. Palaeomagnetic directions from sites in the Bjerkreim-Sokndal layered intrusion, Rogaland Igneous Complex.

Site Number	Rock Unit	*N*	*X*	$\underline{J}$ (A m^{-1})	INC	DEC	*K*	a_{95}	POLE LAT	POLE LONG
BK 1	MCU IV f	4	0.117355	4.45	− 61.9	295.2	62	10.2	− 24.9	232.9
BK 2	MCU IV f	4	0.073573	1.99	− 52.6	309.2	116	7.4	− 11.0	227.5
BK 3	MCU IV e	7	0.109143	3.68	− 66.2	335.8	110	5.8	− 18.9	203.1
BK 4A	MCU IV c	4	0.009354	2.39	− 69.4	303.1	88	9.8	− 31.9	222.4
BK 4B	MCU IV d	2	0.154000	5.82	− 58.7	324.4	228	11.7	− 13.5	217.4
BK 5	MCU IB c	5	0.009185	1.42	− 42.3	309.6	48	10.0	− 2.9	230.7
BK 7	MCU IB e	6	0.066768	2.92	− 69.7	346.2	31	12.1	− 24.0	194.9
BK 8	MCU II c	5	0.002371	1.11	− 67.0	296.9	425	3.7	− 29.8	228.2
BK 10	MCU IA a	5	0.014021	1.29	− 41.9	336.4	145	6.4	3.7	207.5
BK 11	MCU IA a	8	0.005704	0.77	− 63.4	327.8	77	6.4	− 16.8	209.5
BK 12	MCU IB a	4	0.003135	1.79	− 52.3	354.3	67	9.8	− 1.6	190.9
BK 13	MCU II a	4	0.076046	4.71	− 70.6	284.8	89	9.8	− 38.4	231.5
BK 14	MCU III c	6	0.004719	0.76	− 65.3	296.1	57	8.9	− 28.2	230.0
BK 16	MCU II c	5	0.002339	1.44	− 57.7	294.9	331	4.2	− 20.9	235.9
BK 20	MCU IV e′	6	0.070125	9.57	− 75.6	69.4	62	8.6	− 42.3	150.9
BK 31	MCU IV a	5	0.001950	1.67	− 70.5	266.1	89	8.1	− 45.8	242.0
BK 32	MCU III a	4	0.015098	2.81	− 79.0	290.8	398	4.6	− 48.0	216.3
BK 34	MCU IV e′	8	0.076076	16.20	− 81.3	339.0	238	3.6	− 42.2	194.5
BK 38	MCU IV e′	9	0.070316	31.40	− 85.5	114.9	37	8.6	− 61.2	169.1
BK 39	MCU IV e	6	0.130207	4.68	− 59.5	313.4	58	8.9	− 16.2	221.5
BK 40	MCU IV b	5	0.024626	0.49	− 79.1	293.1	106	7.5	− 46.2	214.9
BK 41	MCU IV e	6	0.112640	4.30	− 48.8	302.1	63	8.5	− 10.4	234.7
BK 42	MCU IV e′	5	0.067745	39.90	− 79.7	346.1	579	3.2	− 40.4	192.2
BK 43	MCU IV f	6	0.087617	1.29	− 74.4	356.6	142	5.6	− 30.8	187.9
BK 44	MCU III d	8	0.116076	3.77	− 70.1	195.2	47	9.9	− 80.6	297.3
BK 45	MCU II c	6	0.004691	0.41	− 78.1	339.2	814	2.4	− 36.6	196.2
BK 46	MCU IV d	6	0.038770	1.13	− 61.5	279.1	98	6.8	− 31.1	244.3
BK 47	MCU III d	5	0.159571	4.45	− 51.1	262.2	98	7.8	− 30.6	264.4
BK 48	MCU III a	6	0.002175	1.11	− 77.3	225.8	93	7.0	− 68.1	238.2
BK 62	Mng	3	0.132903	3.84	− 69.7	333.2	59	16.2	− 24.3	203.4
BK 66	MCU IV d	6	0.047206	2.62	− 82.4	354.0	159	5.3	− 43.7	188.4
Bk 67	MCU IV c	6	0.004512	1.62	− 74.4	260.3	61	7.9	− 51.9	237.4
BK 73	QzMng	5	0.024844	0.15	− 74.1	283.7	42	11.9	− 43.9	227.9
BK 74	MCU IV e′	5	0.073957	1.86	− 68.1	30.5	243	4.9	− 23.9	165.7
BK 75	MCU III c	7	0.004477	1.33	− 83.6	336.7	65	7.5	− 46.6	193.6
BK 76	MCU IV e′	5	0.083312	3.36	− 60.5	310.2	68	9.3	− 18.2	223.4
BK 77	MCU IV f	5	0.080430	0.52	− 61.6	281.2	86	7.4	− 30.3	242.7
BK 78	MCU III e	7	0.131857	4.08	− 73.6	27.3	70	7.3	− 29.9	170.7
BK 83	MCU IV c	3	0.006970	2.66	− 64.9	253.8	57	13.4	− 46.2	257.7
BK 84	TZ	4	0.048100	0.77	− 73.0	294.6	59	12.0	− 39.1	223.6
BK 86	MCU IV b	5	0.018922	0.55	− 68.1	276.7	84	7.5	− 38.8	237.1
BK 88	MCU IV d	6	0.024600	6.82	− 51.0	309.5	363	3.2	− 9.4	228.1
BK 90	MCU III b	5	0.001545	1.00	− 80.5	313.9	456	3.2	− 43.9	204.7
BK 91	MCU III c	5	0.001348	0.68	− 71.0	281.4	141	5.8	− 40.1	232.7
BK 92	TZ	5	0.058110	2.33	− 71.4	267.9	20	17.4	− 46.8	240.6
BK 94	Mng	3	0.042997	0.59	− 67.6	280.3	518	5.4	− 36.8	237.6
BK 95	MCU IV e′	5	0.066077	7.19	− 82.2	223.4	102	6.8	− 67.4	214.4
BK 96	MCU IV e′	6	0.068607	9.88	− 77.1	183.4	408	3.3	− 84.3	200.2
BK 97	MCU IV e′	7	0.063548	14.38	− 83.5	320.1	256	3.8	− 49.4	198.7
BK 98	MCU IV e′	5	0.059507	28.21	− 78.7	307.9	186	6.6	− 43.7	209.9
BK 99	MCU IV e′	5	0.040052	29.27	− 70.6	271.0	55	9.3	− 43.8	239.3
BK 100	MCU IV e′	6	0.017356	19.73	− 76.0	263.8	70	7.4	− 52.0	232.5
BK 101	MCU IV e′	4	0.076002	55.01	− 75.5	11.0	106	7.7	− 31.5	180.5
BK 102	MCU IA c	5	0.002501	2.47	− 55.7	314.1	359	3.6	− 12.2	222.4
BK 103	MCU IV c	5	0.002627	0.88	− 83.0	273.6	176	5.2	− 55.2	210.9
BK 104	MCU IV e′	5	0.066580	3.21	− 59.1	313.8	399	3.4	− 15.7	221.3
BK 105	MCU IV e′	6	0.060206	8.73	− 58.9	350.5	108	5.9	− 8.5	193.6
BK 106	MCU III a	7	0.003198	1.39	− 85.9	79.8	401	2.8	− 56.2	171.8
BK 107	MCU IV e′	6	0.035771	18.66	− 76.4	350.9	97	6.3	− 32.9	191.0
BK 110	MCU IA a	5	0.004501	1.99	− 65.6	263.8	170	5.9	− 42.7	251.2
BK 111	MCU IB a	5	0.004204	1.71	− 69.5	247.0	275	4.6	− 54.2	256.3
BK 112	MCU IV e′	5	0.067241	59.74	− 79.1	249.3	203	5.4	− 59.6	227.9
BK 113	MCU IV e′	5	0.058904	57.20	− 80.5	261.6	85	8.3	− 56.4	220.9
BK 114	MCU IV e′	5	0.051947	50.81	− 60.5	283.1	60	9.9	− 28.4	242.4
BK 115	MCU IV c	6	0.011068	2.73	− 64.0	340.1	246	4.3	− 15.4	200.6
BK 116	MCU IV c	6	0.015871	4.30	− 73.8	244.9	145	5.6	− 58.1	245.8

Rock Unit, megacyclic unit, as listed in Fig. 3; *N*, number of samples used in calculations; *X*, susceptibility in SI units; *J*, average initial intensity, in A m^{-1}; INC, inclination, in degrees; DEC, declination, in degrees; *K*, precision parameter (Fisher 1953); a_{95}, cone of confidence around the mean, in degrees; POLE LAT, POLE LONG, latitude and longitude of magnetic pole, in degrees.

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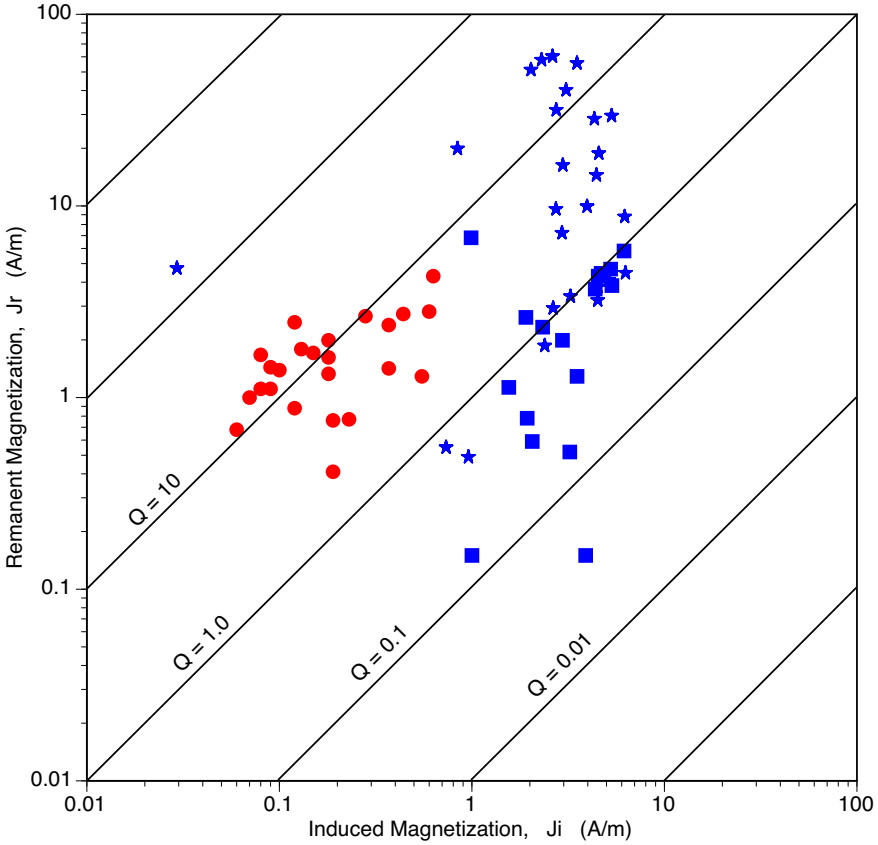


Figure 7. Logarithmic plot of mean induced magnetization (Ji) versus mean remanent magnetization (Jr) for each of the sites from the BKS. Symbols are keyed to oxide composition with red circles = hemo-ilmenite only rocks, blue squares = magnetite only rocks, and blue stars = rocks with both hemo-ilmenite and magnetite present. A value of 39.814 A m^{-1} was used for the present field strength in Rogaland.

Hysteresis properties of 143 BKS samples, representing the 66 sites studied here, show strong correlation to magnetic mineralogy (Fig. 8), as would be expected from previous work by McEnroe *et al.* (2009). The magnetite-bearing rocks (zones IV d through f and the capping mangerites) where the co-existing ilmenite does not contain exsolution lamellae of hematite have a range of remanent magnetization to saturation magnetization ratios (Mr/Ms) and high coercivity of remanence to coercivity ratios (Hcr/Hc) that fall along a continuum from the lower portion of the PSD area into the MD field. Seventy-six percent of these samples have Mr/Ms ratios below 0.05 indicating the presence of large multi-domain magnetite grains as observed in thin section. The magnetite grains from this part of the intrusion are millimetre in size and commonly have limited microstructures, except for samples from the transition zone (TZ). The TZ samples (Fig. 6a) have significantly higher Mr/Ms ratios (1.4–1.2) and plot well into the PSD region. These samples all have fine exsolution microstructures in the magnetite due to the early exsolution of ulvöspinel (100) from the initial titanomagnetite (Fig. 6a). Samples with hemo-ilmenite, and rare (<0.05 per cent), or no magnetite, show a wide range of Hcr/Hc values from 1.5 to 9, and higher Mr/Ms values ranging from 0.09 to 0.4. These samples lie above the MD magnetite line of 0.05 Mr/Ms with half having Hcr/Hc ratios above 4. These high Hcr/Hc ratios could be due to the nanoscale hematite lamellae within the paramagnetic ilmenite, and thus be superparamagnetic, because the nanoscale lamellae would not have a stable coupling above the ordering temperature of ilmenite (57 K). In contrast hematite lamellae >30 nm provide a stable coupling environment for the ilmenite lamellae and thus will contribute a strong lamellar moment. This compares to the behaviour of the ilmeno-hematite samples observed by McEnroe & Brown (2000) from the Adirondack Mountains, where the nanoscale ilmenite lamellae and associated contact moment is coupled to the hematite host, thus providing a stable contact moment (Robinson *et al.* 2002, 2004). This high-coercivity moment is also observed in samples of the near-by Egersund-Ogna anorthosite (Brown & McEnroe 2004) that contain inclusions of ilmeno-hematite. Mixed mineralogy samples of magnetite (Figs 6c and d) and hemo-ilmenite show a similar trend to the magnetite only samples, reflecting the large contributions of MD magnetite at the high fields (1–2 T) used in the typical hysteresis measurements, which will mask the presence of hemo-ilmenite.

4.4 Palaeomagnetic behaviour

Both progressive thermal and alternating field demagnetization reveals stable magnetic directions for many of the BKS samples. Detailed demagnetization behaviour can be divided into three groups, corresponding to the observed variations in mineralogy: (I) Hemo-ilmenite dominated samples which show little decay in AF demagnetization up to 100 mT and thermal demagnetization with little loss in intensity until temperatures above 575 °C. (II) Mixed mineralogy samples with commonly large NRM values, variable AF behaviour, and thermal components persisting to above 575 °C. (III) Magnetite-only samples with soft AF demagnetization behaviour and thermal demagnetization that falls off at, or before 550 °C.

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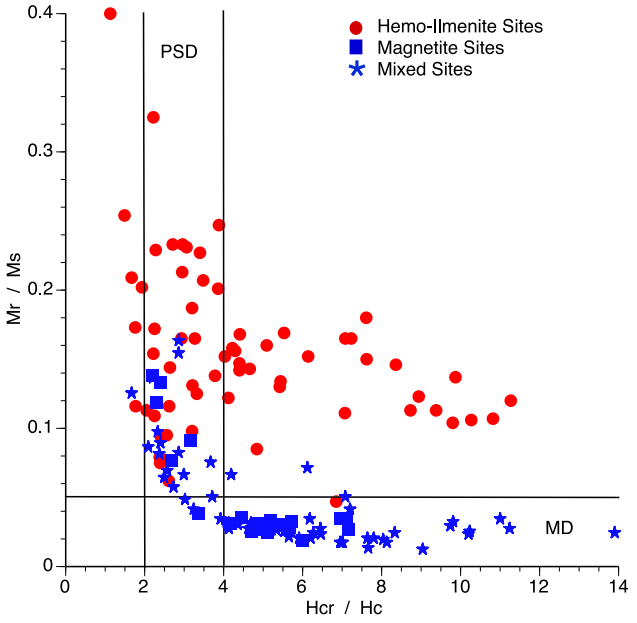


Figure 8. Plot of ratio of remanent magnetization to saturation magnetization (Mr/Ms) to the ratio of the coercivity of remanence to the coercivity (Hcr/Hc) for selected samples from the BKS. Symbols indicate oxides present, as in Fig. 7. Lines separating pseudo-single domain (PSD) from multidomain (MD) are from Day *et al.* (1977) as modified by Dunlop & Özdemir (1997). Note boundaries of these fields pertain to magnetite-bearing rocks only; additionally hemo-ilmenite samples may have not saturated and thus do not represent a true Mr/Ms value.

Samples with behaviour I are found in the lower MCU sections of the cumulate layers, and in the more primitive recharge zones described as 'a' or 'c' (Fig. 3). These rocks have magnetic properties controlled entirely by hemo-ilmenite. Alternating field demagnetization has little influence on intensity, with mean destructive fields (MDFs) of this group are commonly greater than 100 mT and typically 40–80 per cent intensity remaining (Fig. 9a). Directions are steep and up with SW to NW declinations, and are stable from NRM to 100 mT (Figs 10a, c and e). Thermal demagnetization is marked by only slight direction change, and little if any decrease in intensity from NRM to 575 °C (Fig. 9b). Directions are consistently steep negative inclinations with declinations from the southwest to northwest (Figs 10b, d and f). By 600 °C intensity has dropped considerably, although directions are still consistent. By 625 °C and above intensity is below 5 per cent of the original. Thermal and AF demagnetization on samples from the same site yield near identical directions.

Group II contains sites that show a range of behaviours on demagnetization, but provide stable, consistent directions. Intensity decay can be quite variable, with both the presence of hard or soft components. Thermal demagnetization can be consistent to 550 or 575 °C or decay slowly from 200 to 600 °C, or even show a systematic increase up to 550 °C (Fig. 9d). AF demagnetization shows a range of MDFs from 5 mT to >100 mT (Fig. 9c). Stable directions are similar to those from Groups I, dominated by steep negative inclinations and SW to NW declinations (Fig. 11). Those with low MDF and high-magnetite content consistently show a high-field component (Fig. 11a) with NW directions reflecting the hemo-ilmenite contribution.

A subset of this group is characterized by large NRM values, commonly greater than 20 A m^{-1} (Figs 11e and f). Intensity determinations on the cryogenic magnetometer are unreliable at this very high level of magnetization and NRM measurements must be considered minimums. Intensities measured on the JR6 are more accurate and strongly magnetic samples were measured at NGU, where maximum NRM values of 60 A m^{-1} were obtained (Table 1). The most strongly magnetized sites come from the Heskestad region in the southeast corner of the Bjerkreim lobe where they are associated with a large negative magnetic anomaly, which can be traced for over 20 Km to the north, and have been described in detail by McEnroe *et al.* (2004a,b). The sites with this behaviour coupled with ground magnetic surveys, have been used to define a further stratification in the intrusion with the introduction of MCU IVe′ within MCU IVe, which, using ground magnetic surveys, has been traced north and westward for over 10 km in the intrusion.

Samples that come from the upper part of MCU IV (zones e and f) and transition zone, mangerite, and quartz mangerite sections on the top of the intrusion show group III demagnetization behaviour. In these units magnetite is prevalent, and commonly multi-domain in size. Intensity tends to fall off rapidly in AF demagnetization, with all samples having MDFs less than 30 mT, and most samples showing MDFs less than 10 mT (Fig. 9e). In some samples intensity decreases slowly during thermal demagnetization until 550 °C after which it drops rapidly (Fig. 9f), while in others it remains high to 500 °C. Samples in this group display a range of overprints; most are removed by 400–500 °C for thermal studies with smaller overprints seen in AF demagnetization studies (Fig. 12). As with the previous groups, characteristic directions usually are steep and up, with SW to NW declinations.

MD magnetite samples, as identified by hysteresis properties (Fig. 8), predominantly show unstable magnetic directions during progressive steps using both AF and thermal techniques. Bulk MDFs in AF demagnetization are very low for these samples, usually less than 5 mT (Fig. 9e). Intensity during thermal demagnetization decreases rapidly in some samples, with 50 per cent removed by 300 °C, while in others intensity is preserved to above 500 °C (Fig. 9f). Initial NRM values are 3 A m^{-1} or less, however the rapid loss of intensity leaves little to be measured at higher demagnetization levels. Inclinations are variable between positive and negative, and declinations show considerable scatter. This behaviour is limited to the mangerite and quartz mangerite samples. Of the six sites from this region acceptable directions could be gleaned from only three of them and the remaining sites yielded scattered directions without meaningful statistical results.

4.5 Metamorphic country rocks

In an effort to better understand the magnetic signature of the BKS we collected nine sites (37 individual cores) from the surrounding metamorphic terrain (Fig. 1). These rocks, accreted to the Baltica margin ~1.5 Ga (Bingen *et al.* 2008a), were metamorphosed to granulite facies at the tail end of the Sveconorwegian orogeny (Bingen & Van Breemen 1998; Bingen & Stein 2003). At the time of RIC intrusion additional high temperature contact metamorphism occurs as far as 20 km from the contact with the intrusive rocks (Westphal *et al.* 2003).

Susceptibility of the metamorphic rocks range over four orders of magnitude, from 3.60×10^{-5} to 4.26×10^{-1}, with an average of 3.80×10^{-2} SI. Correspondingly, the NRM values of these samples vary greatly, from 0.33 to 6.10 A m^{-1}. On AF and thermal demagnetization many of samples yield present field overprints and

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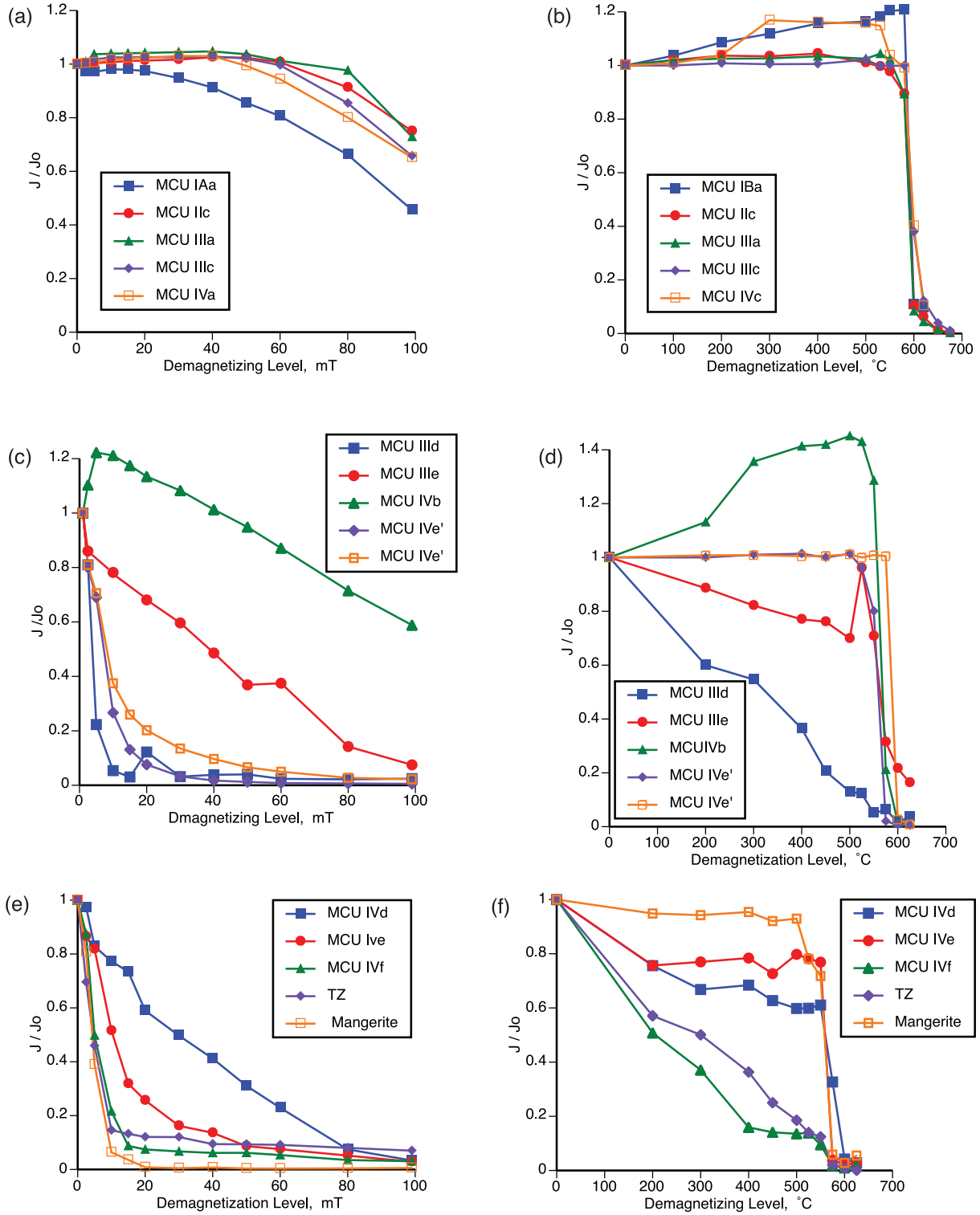


Figure 9. Plots of remanence normalized over demagnetization steps (J/Jo). Left-hand column, alternating field demagnetization from 0 to 100 mT; right-hand column, thermal demagnetization from 0 to 700 °C. (a, b) Samples from hemo-ilmenite dominated rocks. (c, d) Samples with both hemo-ilmenite and magnetite present. (e, f) Samples with magnetite as the only magnetic phase.

scattered directions. These sites rarely showed consistent directions with acceptable statistics. However, a subset of samples from all but one site showed early Neoproterozoic directions. A mean for 20/37 selected samples from eight sites, shown in Fig. 13, is $I = -65.8^\circ$ and $D = 294.1^\circ$ with $\alpha_{95} = 10.8^\circ$, consistent with directions from the BKS. Due to the large scatter and small number of samples this direction should not be used as a pole position, and only is shown to indicate the remagnetization of the country rocks by the intrusion of the very hot 7 km thick BKS and the associated anorthosites into the surrounding metamorphic rocks. All eight sites in the

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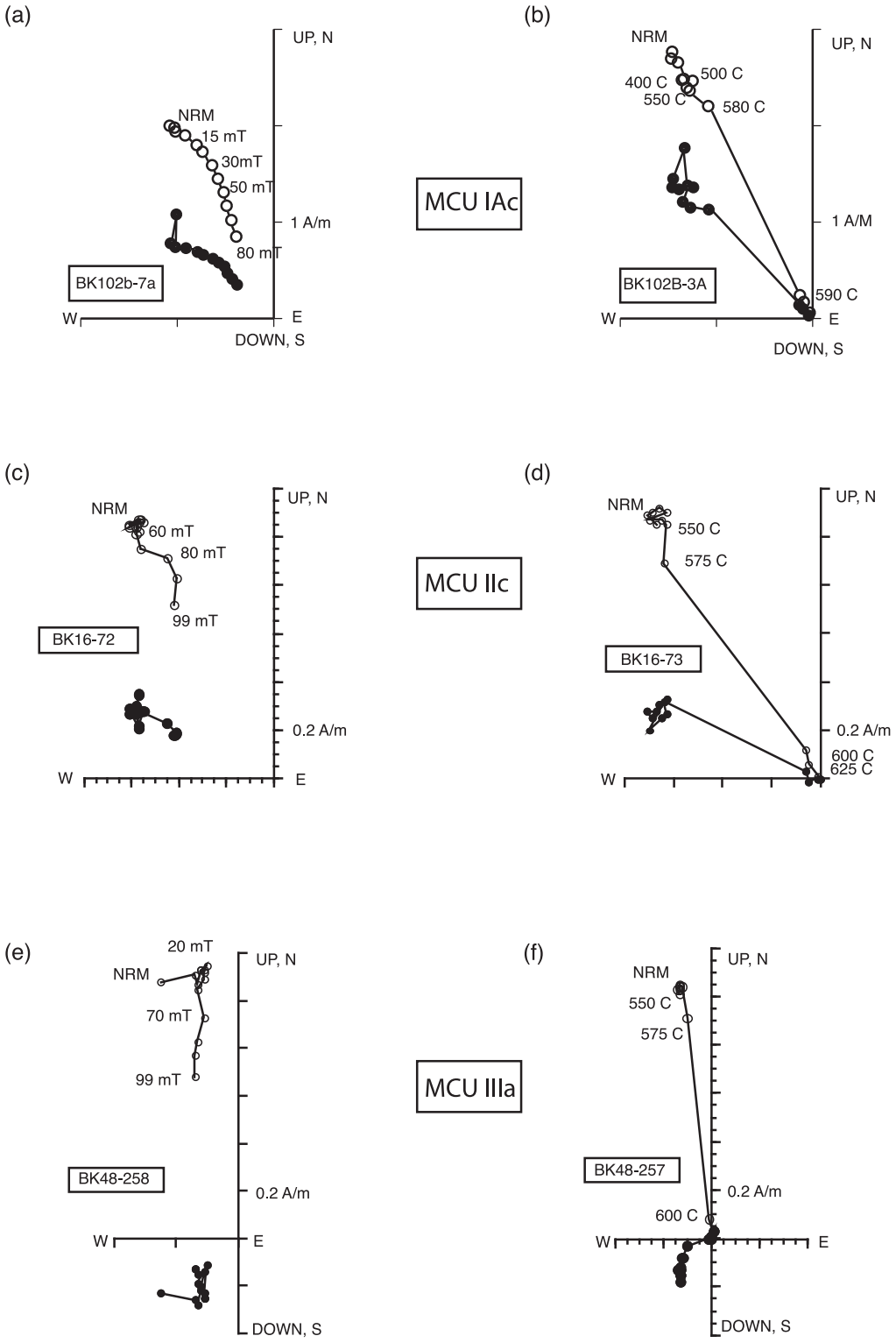


Figure 10. Vector end-point diagrams for alternating field (left column) and thermal (right column) demagnetization. All samples come from the lower part of megacyclic units where hemo-ilmenite is the dominant remanence carrier. Open circles are projections on the vertical plane; solid circles represent projections on the horizontal plane.

country rock lie within the large metamorphic aureole associated with the intrusion and thus no remanent magnetization directions were obtained in the country rock outside of the influence of these later intrusions, making a classic positive contact test (e.g. Buchan 2013) impossible.

5 RESULTS

Palaeomagnetic data for 66 sites from the BKS are listed in Table 1 with magnetic directions plotted in Fig. 14(a). The mean direction of this data set ($N = 66$) is inclination $= -73.5^{\circ}$,

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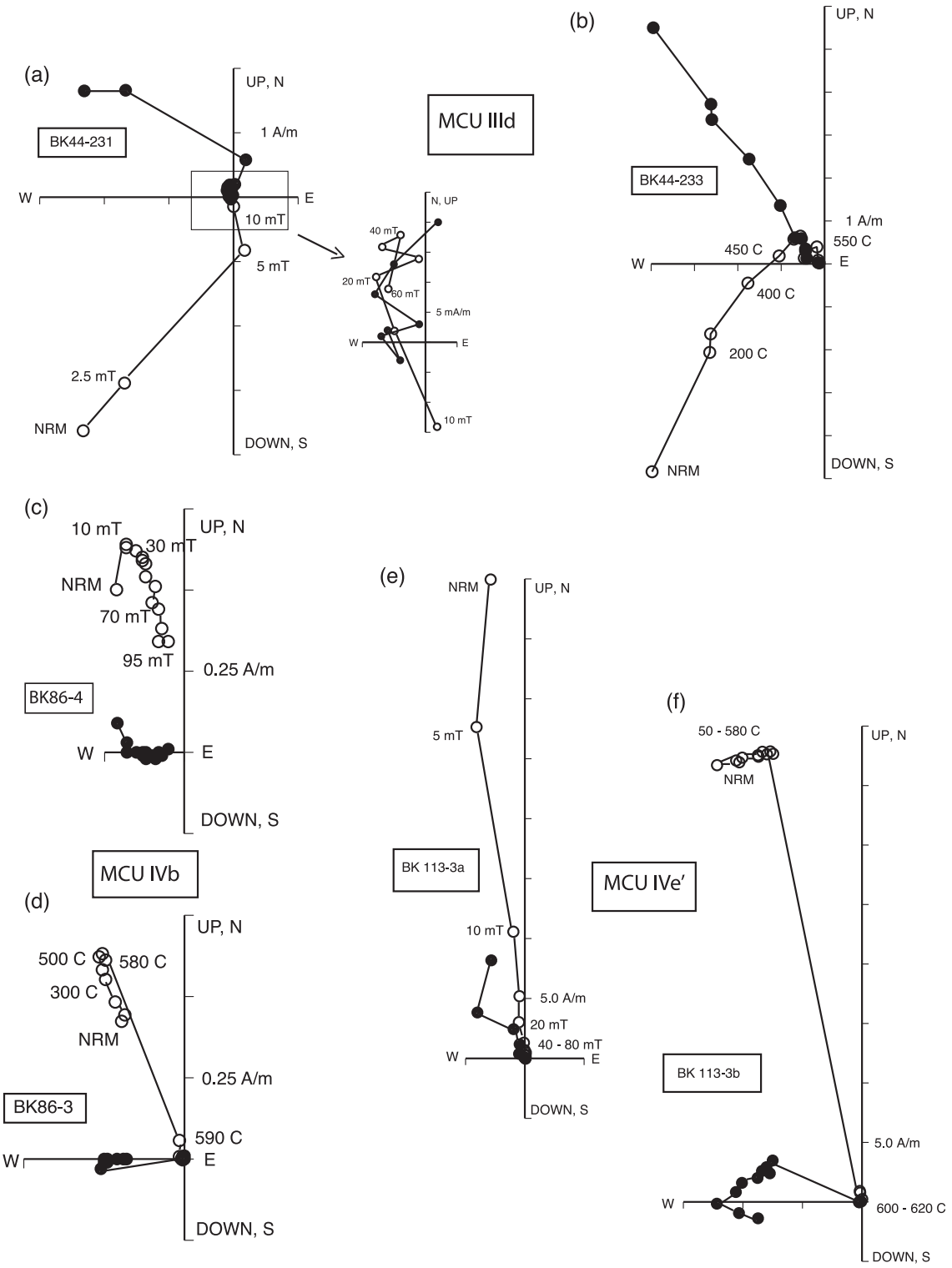


Figure 11. Vector end-point diagrams for alternating field (left column) and thermal (right column) demagnetization. All samples come from parts of megacyclic units where both hemo-ilmenite and magnetite contribute to the remanence. Enlargement of part of (a) shows persistence of NW directions even in samples with high magnetite content. Symbols as in Fig. 10.

declination = 303.4°, with $k = 24$ and $\alpha_{95} = 3.7°$. A corresponding palaeomagnetic pole is found at 35.9°S and 217.9°E. Results from the different MCU units in the body, representing different stratigraphic levels in the intrusion, yield statistically similar directions (Table 2, Fig. 14b). When we separate the data into the different magnetic mineralogy; hemo-ilmenite-dominated rocks (subunits a, b, and c), magnetite only rocks (subunits e, and f in MCUIV and the overlying norite, mangerite and quartz mangerite), and mixed (hemo-ilmenite and magnetite), the mean directions overlap at the 95 per cent confidence level (Table 2;

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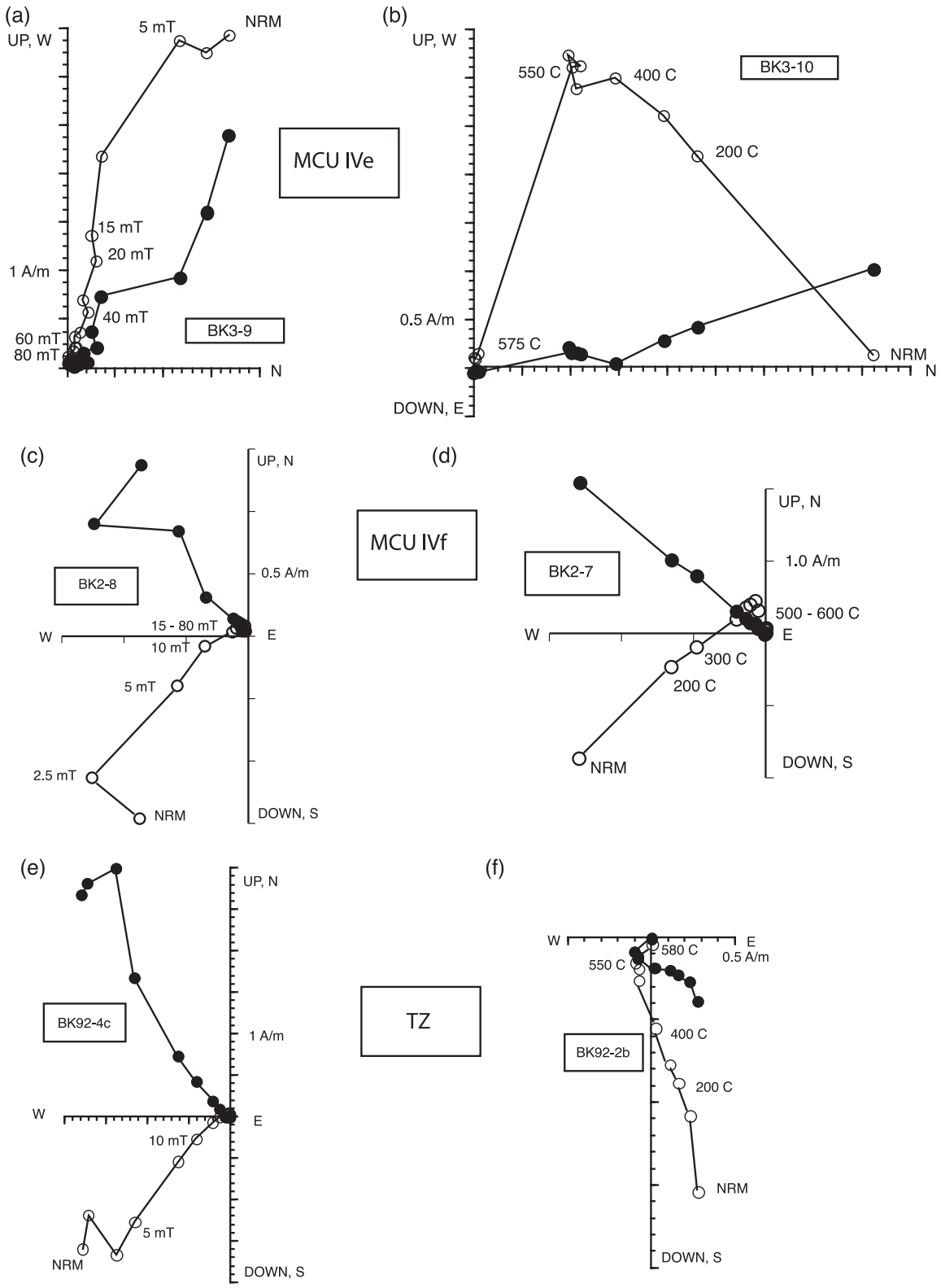


Figure 12. Vector end-point diagrams for alternating field (left column) and thermal (right column) demagnetization. All samples come from the upper parts of megacyclic units and the overlying transition zone where magnetite is the dominant remanence carrier. Symbols as in Fig. 10.

Fig. 14c). Additionally, when further subdivisions are considered, and directions averaged for individual subunits within the MCUs (Table 2), there is no consistent difference seen in units throughout the body.

6 DISCUSSION

Sixty-six sites from the BKS provide stable palaeomagnetic directions and corresponding virtual magnetic poles. Although the carriers of the remanence vary from hemo-ilmenite, or magnetite,

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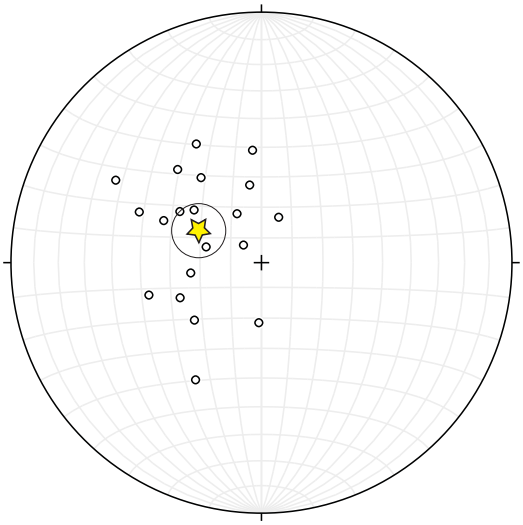


Figure 13. Stereographic plot of selected samples from the metamorphic rocks north and west of the BKS, representing eight of the nine sites collected. Yellow star is mean inclination and declination of the 20 samples portrayed, with its circle of 95 per cent confidence.

or a combination of both minerals, the directions recorded are similar. The hemo-ilmenite dominated sites, coming from MCU subunits a and c are the most stable, with no overprints and identical directions from both AF and thermal demagnetization methods. These sites are noted for low susceptibility values, and with NRM values commonly reach several A/m. Sites with both hemo-ilmenite and magnetite present show solid magnetic behaviour and stable directions, with the MCU IVe′ sites yielding anomalously strong, but stable directions. In the upper MCU units magnetite increases in size (>500 μm) and abundance, while at the same time the Fe^{3+} component in hemo-ilmenite decreases in abundance, which results in a less stable remanence. By the time the mangerites and quartz mangerites appear at the top of the intrusion the magnetite is very coarse-grained with fewer microstructures, and many of the samples prove to be dominated by soft components, and commonly yield scattered directions.

We find no evidence that a widespread alteration to the silicate or oxide mineralogy in the BKS has occurred. This has been verified by our extensive silicate and oxide petrography, and petrological and geochemical studies by others, as discussed earlier. The seven km thick BKS has a primary magnetization held in a combination of magnetic minerals, both hemo-ilmenite and magnetite, which record similar magnetic directions obtained at the time of cooling through their respective blocking temperatures.

6.1 Age of remanence

After detailed laboratory studies and once the reliability of the remanence is established, it is necessary to consider the age of remanence. In the case of a large (5000 km^2) intrusion such as the BKS, the time when it cools below the blocking temperatures of the oxide phases will be younger than the U-Pb ages determined for the body. As discussed for metamorphic and igneous units in the Adirondack Mountains cooling from an orogenic event, magnetites and ilmeno-hematites, depending on composition, can obtain their remanence at different times, and these ages can be determined from a detailed cooling curve (Brown & McEnroe 2012). In the situation here, we must consider the cooling of the BKS from the closure temperature for the U-Pb system of zircons (∼850 °C) at 931 Ma down

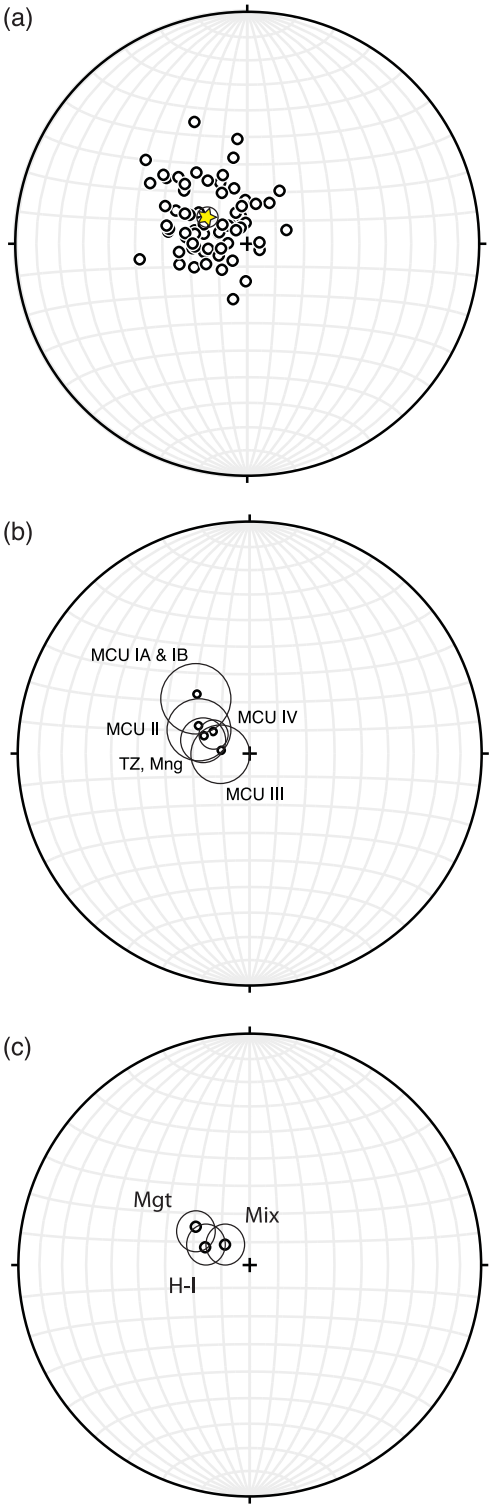


Figure 14. Stereographic plots of mean directions from the BKS. (a) All individual sites; Star is mean direction for all 66 sites, surrounded by its circle of 95 per cent confidence. See Table 1 for data. (b) Mean directions for the four megacyclic units plus the upper transition zone, mangerites and quartz mangerites, with their circles of 95 per cent confidence. See Table 2 for data. (c) Mean directions of sites categorized by oxide composition with 95 per cent circles. H-I, hemo-ilmenite only samples; Mix, samples with both hemo-ilmenite and magnetite phases; Mgt, samples with only magnetite present. See Table 2 for data.

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Table 2. Mean magnetic directions and poles for MCU units, subunits, upper mangerite zone and different oxide mineralogies for the Bjerkreim-Sokndal layered intrusion.

UNIT	SITES	INC	DEC	α_{95}	POLE LAT.	POLE LONG.
TZ, Mng, Qtz	5	− 72.5	291.8	8.0	− 38.2	225.2
IV f	4	− 64.6	304.4	17.4	− 24.5	224.9
IV e	3	− 58.9	314.2	11.4	− 15.3	221.1
IV e′	19	− 81.6	322.3	6.3	− 44.5	200.3
IV d	4	− 64.8	307.4	19.5	− 23.8	222.7
IV c	6	− 74.3	274	8.5	− 46.5	231.5
IV b	2	− 73.7	282.2	26.2	− 42.9	228.5
IV a	1	− 70.5	266.1		− 45.8	241.9
IV w/o e′	20	− 68.9	296.4	5.5	− 32.2	226.5
IV all	39	− 74.7	301.4	4.6	− 38.1	217.6
III e	1	− 73.6	27.3		− 30.0	170.6
III d	2	− 64.4	239.8		− 52.9	268.5
III c	3	− 74	295.4	17.1	− 38.9	221.5
III b	1	− 80.5	313.9		− 43.9	204.7
III a	3	− 84.7	254.7	15.9	− 59.7	206.6
III all	10	− 79.7	277.7	10.1	− 51.0	218.8
II c	3	− 68.4	303.3		− 29.4	222.7
II a	1	− 70.6	284.8		− 38.3	231.4
II all	4	− 69.2	298.8	12.5	− 31.8	224.8
I B	4	− 63.7	321	29.3	− 18.7	214.1
I A	4	− 59.4	316.2	22.3	− 15.2	219.4
I all	8	− 61.6	318.4	14.5	− 16.9	216.8
Hemo-il only	23	− 72.8	299.2	5.9	− 38.4	224.7
Mixed Min	27	− 78.5	309.7	6.3	− 41.5	209.2
Mgt only	16	− 66.3	305.0	6.2	− 26.3	223.3
All BKS sites	66	− 73.5	303.4	3.7	− 35.9	217.9

UNIT refers to stratigraphic divisions as in Fig. 3. Other designations as in Table 1.

to blocking temperatures for both magnetite and hemo-ilmenite mineralogies. Previous work by Bingen & van Breemen (1996) determined a regional cooling rate of 15 °C myr^{-1} for the RIC (obtained from U/Pb titanite ages (∼600 °C closure temperature) in the contact aureole of the BKS dated at 918 ± 2 Ma). This agrees well with ^{40}Ar/^{39}Ar hornblende ages (∼530 °C closure temperature) of 916 +12/−14 Ma, also obtained in gneisses within the contact aureole (Bingen *et al.* 1998b). The blocking temperatures for the hemo-ilmenite compositions in the BKS are very limited, and the magnetization should be acquired slightly below 520 °C. Magnetite should block by 550–570 °C, depending on composition. Given the modelled rate for cooling of 15 °C myr^{-1} the difference in the time of remanence acquisition for different mineralogies is not large. In fact, we would expect more variation in the timing of blocking due to location of samples within the 7 km thick intrusion, than amongst the different blocking temperatures of the various mineral phases. At the time of emplacement sites closest to the cooler metamorphic rocks would have cooled earlier than those at the centre or deeper in the intrusion. Magnetic directions at the 95 per cent level are similar for MCUs (Figs 14b and c), indicating either the phases locked in their remanence over a relatively short time period and/or magnetic directions in southern Baltica were not changing rapidly, at this time, due to either changes in the Earth's field, or rapid plate motions. Thus, an age of 916 Ma, given the emplacement age, the cooling rate, and the titanite and argon ages, is a sound estimate for acquisition of remanence for the entire BKS intrusion.

6.2 Primary remanence

Several recent papers (Walderhaug *et al.* 1999, 2007) suggest that all the rocks in the RIC may have had their magnetizations reset around 850 Ma. The evidence presented for this assumption is based on the magnetic directions held by four Hunnedalen dikes (six total sites), intruding the Sveconorwegian granulite terrain northwest of Rogaland, the ages of the dikes, and the similarity of the magnetic directions to those in the RIC. Walderhaug *et al.* (1999) present two isotopic measurements for the dikes of 855 ± 59 Ma by Sm-Nd whole rock-mineral system and 848 ± 27 Ma from ^{40}Ar/^{39}Ar plateau age on biotite. The Hunnedalen dikes, as described by Maijer & Verschure (1998), have very similar chemical compositions to the monzonorites (jotunites) of the RIC, and are suggested to have generic ties to the 930 Ma igneous complex. It is possible that the age determinations by Walderhaug *et al.* (1999) have been influenced by low-grade alterations in that part of Rogaland, and may not reflect a primary age of the magnetization of these dikes.

The average pole for the six Hunnedalen dike sites (41°S, 222°E, A_{95} = 10°), referred to as a VGP by Walderhaug *et al.* (1999), is close to the mean pole obtained here for the BKS, as well as other units within the RIC (Table 3). There is no indication in the petrology or mineralogy of the BKS or in the magnetic behaviour of the BKS samples that they have been subjected to any 'short lived metamorphic or hydrothermal event' as suggested by Walderhaug *et al.* (1999). The hemo-ilmenite-dominated samples from subunits a, b and c show no overprints or secondary directions (Figs 10a–f), or give any other indication that the directions are not primary. Some magnetite-dominated samples exhibit components removed at temperatures below 400 °C (Fig. 12b), but directions at high temperature are well clustered. To reset both the magnetite and hemo-ilmenite carriers in the seven km thick BKS body would require elevated temperatures and/or massive fluid alterations visible in the total rock composition. Despite extensive mineralogical, petrological, and geochemical investigations of these rocks

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Table 3. Palaeomagnetic Poles from southern Scandinavia.

Study	Age (Ma)	Sites	Pole Lat	Pole Long	A_{95}	Des	Reference
Bamble Intrusions	∼1070	4[a]	−3	37	15	I	Meert & Torsvik (2003)
Dikes, South Sweden	935–946	11	−0.5	241	6	S	Elming *et al.* (2014)
Bamble Terrane	950–900	(191)	−32	231	3	B	Piper (2009)
Telemark Terrane	950–900	(151)	−34	239	4	T	Piper (2009)
BKS	916	66	−36	218	6	Star	This paper
Hakefjorden Complex	916 ± 11*	11	5	249	4	F	Stearn & Piper (1984); Scherstén *et al.* (2000)
Egersund Ogna	∼900	13	−42	200	9	E	Brown & McEnroe (2004)
RIC	940–900	76	−44	214	4	R	Stearn & Piper (1984)
Hunnedalen Dikes	848 ± 27 855 ± 59	6	−41	222	11	H	Walderhaug *et al.* (1999)

Age of remanence as reported in original publication; *radiometric age determined after palaeomagnetic data published; A_{95}, cone of confidence at 95 per cent level around pole; Des, study designation on Fig. 13; [a]number of studies combined; ( ), number of components (not samples or sites), as reported by Piper (2009).

(Michot 1960; Wilson *et al.* 1996; Duchesne 2001; McEnroe *et al.* 2009; and others) there is no evidence of a later metamorphic, hydrothermal, or chemical event of the calibre necessary to reset all the magnetic directions observed here in this massive intrusion. Not only are the majority of the oxides pristine, but also the olivine, a mineral notorious for altering in presence of a small amount of fluid, is in excellent condition (Fig. 4). In all of our demagnetization experiments secondary magnetic directions from either the magnetite rocks or the magnetite + hemo-ilmenite rocks are removed at relatively low fields. Samples from sites with only hemo-ilmenite have high coercivities and do not show secondary magnetizations. Extensive mineralogical, geochemical and petrologic investigations have revealed most samples to be in excellent condition, with little or no alteration. In addition, the surrounding country rock was remagnetized by the intrusion of the BKS, and evidence of this high temperature metamorphism can been seen in the adjoining rocks (Westphal *et al.* 2003). We believe, based on extensive mineralogical and petrologic data, the palaeomagnetic directions reported here are of a primary nature, and represent the magnetic field present at ∼916 Ma. Although only one polarity is found for the BKS, other units in the RIC, the contact aureole around the RIC and the Hunnedalen dikes, this may only be an indication of a variation in reversal frequencies in the Precambrian (Biggin *et al.* 2008). The age of the RIC magnetization is important to establish as other researchers are using the younger Hunnedalen date (∼850 Ma) for results from the RIC (Elming *et al.* 2014).

6.3 Early Neoproterozoic Baltica

Many authors have discussed the position of Baltica during the early Neoproterozoic, and, as with other cratonic pieces, it enjoys different positions in various reconstructions. Early reconstructions of Laurentia and Baltica have the two continents juxtaposed at ∼1.0 Ga, the time of the Grenville and Sveconorwegian orogenies (Patchett *et al.* 1978). Subsequent models for the supercontinent Rodinia continue this relationship, usually with Baltica off-board of northern Laurentia (Greenland) (Hoffman 1991; Dalziel 1997). More recent interpretations continue this 'classic' Grenville connection, with another craton, usually Amazonia, south of both Baltica and Laurentia (Pisarevsky *et al.* 2003; Li *et al.* 2008; and others). Variations on these positions are quite allowable, particularly with the paucity of well-dated and robust palaeomagnetic poles from many of interacting cratons over the life span of Rodinia. Different scenarios include repositioning Amazonia (Pesonen *et al.* 2012), removing it completely from the area (Evans 2009), and reinterpreting the southern Baltica edge as accreting rather than collisional (Slagstad *et al.* 2013). Despite the exact positions and relationships between cratons, and variations of when Rodinia amalgamated and its subsequent dispersal, it appears to be an intact supercontinent at the time of emplacement of the RIC.

Within the Rodinia supercontinent there are variations concerning the rotation of Baltica from it present orientation, with a suggestion by Hartz & Torsvik (2002) that Baltica was actually inverted in its position adjacent to eastern Laurentia by the initial breakup of Rodinia (∼750 Ma). Subsequent considerations of the regional geology and palaeomagnetism prefer Baltica in classic 'upright' position with southeast facing Baltica opposing southwest facing Laurentia (Bingen *et al.* 2008a, 2011; Cawood *et al.* 2010). In addition, Baltica is most commonly placed not only in close proximity to Laurentia, but also near to Amazonia, the assumed continental craton with which southeastern Baltica collided during the Sveconorwegian orogeny (Li *et al.* 2008; Bingen *et al.* 2008b).

Palaeomagnetic data of Early Neoproterozoic age from the southeast corner (southern Scandinavia) Baltica craton are listed in Table 3 and displayed in Fig. 15 where all pole positions have been inverted from Table 3 except the Bamble Intrusions, previous inverted by Meert & Torsvik (2003). Obviously of importance to this discussion is the lack of detailed studies within the Fennoscandia region of Baltica that have both robust palaeomagnetic data and convincing determinations of the age of remanence. We employ criteria in determining which poles to include in this compendium using the criteria of Van der Voo (1990) and adding the requirements of: (1) at least 10 distinct sites to determine a pole position, (2) sites have agreement in thermal and AF directions, and (3) the need for discussion of the age of the unit and its relationship to the age of the remanence. Using the recently compiled data on Precambrian pole studies, PALEOMAGIA, (Veikkolainen *et al.* 2014) for the time period 1000–850 Ma, over 100 entries are listed for Baltica. Many of these are on studies with only a few sites or without established age information, and thus were rejected from further consideration. Additionally a number of units with known ages are given younger remanence ages (850 Ma), assumedly based on the comment by Walderhaug *et al.* (1999) on remagnetization, as discussed above. Recent compilations of Baltica poles for Rodinia constructions are limited to roughly a dozen studies (Li *et al.* 2008; Evans 2009).

Employing the criteria set above, with minor exceptions, discussed below, produces a pole list for this area with nine paleopoles (Table 3; Fig. 15), including three studies from the Rogaland area with remanence ages close to 900–920 Ma. This study, an early reconnaissance study by Stearn & Piper (1984) on many units in the RIC, and a study on the Egersund-Ogna anorthosite

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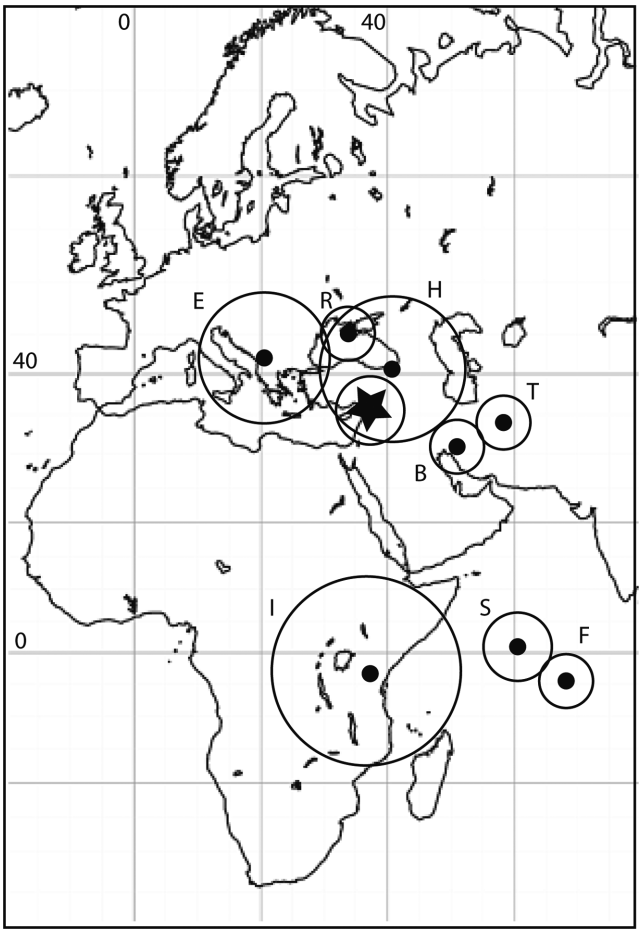


Figure 15. Plot of pole locations for studies in southern Scandinavia shown with A_{95} circles of confidence; selection criteria as discussed in the text. Star, pole for BKS; B, Bamble Terrane; E, Egersund-Ogna; F, Hakefjorden; H, Hunnedalen; I, Bamble Intrusions; R, Rogaland Igneous Complex; S, Swedish dikes; T, Telemark Terrane. Data and sources listed in Table 3.

(Brown & McEnroe 2004) define poles in mid-latitude between 20° and 48°E longitude; although the three studies plot close to each other, the pole from the Egersund-Ogna anorthosite does not overlap with the BKS at the 95 per cent level.

A new compilation of 11 dikes, representing sites across southern Sweden, and with ages of 936–943 Ma, provides a distinctly different pole with an equatorial position (Elming *et al.* 2014); it also is the only pole to satisfy the strict criteria of a 'key pole' (Buchan 2013) by having a positive contact test as well as well-determined rock ages. Two pole positions for rocks from the Bamble and Telemark terranes of eastern Norway (Piper 2009) are included in Table 3, however the age of remanence is not sufficiently constrained to make them reliable. In addition, the directions are determined from components of magnetization and the number of sites involved is not evident. These poles fall in low to mid latitudes, between the RIC poles and the Swedish Dikes pole. Norites from southwestern Sweden, referred to as the Hakefjorden complex, sampled by Stearn & Piper (1984), provide data from 11 sites, although the authors report that five of the sites were collected in 'altered norite'. An experimental dating method using contact-melt zircons has provided a date of 916 ± 6 Ma (Scherstén *et al.* 2000), although this appears to be on a granite back-vein intruded into the palaeomagnetically sampled unit and thus the age relationships are not clear. The pole from this study falls in equatorial latitude near the Swedish dikes pole (Fig. 15).

The youngest rocks in the area currently appear to be the Hunnedalen dikes (Walderhaug *et al.* 1999), although with only six sites representing four dikes, imprecise ages, and their geologic similarity to RIC rocks, they are hard to interpret. These dikes were also studied by Poorter (1972) who sampled six dikes (several the same as sampled above) with a total of eight sites, but the absence of individual site data does not allow the populations to be combined. Because of their location and similar magnetic direction we make an exception to our minimum 10 site criteria and include the data in Table 3. As shown in Fig. 15, the mean poles from the BKS and the Hunnedalen dikes overlap at the 95 per cent level.

Regrettably there are no palaeomagnetic studies of any younger rocks in this area until the 616 ± 3 Ma Egersund dikes (Bingen *et al.* 1998a; Walderhaug *et al.* 2007). Likewise, few poles of older age (1100–950 Ma) exist in this area for comparison; the Bamble intrusions are included in Table 3, representing a compilation of four studies (Meert & Torsvik 2003) but determined from original data collected by Stearn & Piper (1984); once again a precise age determination for either the rocks or the acquisition of remanence is lacking as is a discussion of any influence from the large faults known in the area or the possibility of a contact test. U-Pb ages on other gabbroic intrusions in the Bamble region, as reported by Nijland *et al.* (2014), are greater than 1200 Ma, suggesting this pole may be much older than reported. The pole position for these intrusions, although equatorial in location, is displaced from the poles for southern Sweden.

As can be seen on Fig. 15, there is a considerable spread in pole positions from 950 to 900 Ma, as well as some marked differences in pole latitudes between the RIC in southwestern Norway and dikes and intrusions in southern Sweden. Both the dikes from Sweden, dated between 946–935 Ma, and the Hakefjorden Complex, dated at 916 Ma, yield equatorial pole positions which are 30–40° further south than studies from southwestern Norway. The differences influence the position of Baltica within Rodinia at this time, as the palaeolatitudes of the two groups indicate either at low latitude (~20° for the Swedish dikes) or high latitude (~60° for the BKS) position for southern Baltica. If you consider the ~20 million year age difference between the BKS and the Swedish dikes, very rapid plate motion in the order of 2° of latitude/myr would be required to reconcile their differences, which we consider unlikely. This does not address Hakefjorden results with an age similar to BKS but directions similar to the Swedish dikes. Nor does it resolve the Hunnedalen dikes with directions similar to the BKS but a younger, albeit somewhat unconvincing, age. Reasons for the variation in directions across southern Scandinavia remain enigmatic, and may only be resolved by additional detailed palaeomagnetic and geochronologic studies in both areas.

7 CONCLUSIONS

(1) The BKS layered intrusion, part of the RIC of southern Norway, dated at 932 ± 7 Ma, preserves stable remanent magnetization in all the five MCUs. Similar directions were obtained from baked host rocks up to 20 km from the intrusion.

(2) Remanent directions are carried by both hemo-ilmenite, in the lower layers of each MCU, followed by magnetite-dominated upper layers, with a region of mixed oxide mineralogy of both hemo-ilmenite and magnetite occurrences within specific layers of the MCUs.

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(3) Palaeomagnetic direction from 66 sites, representing all four major magmatic cycles and subcycles, as well as the transition zone and upper mangerites, are $I = -73.5^\circ$ and $D = 303.4^\circ$, with $\alpha_{95} = 3.7^\circ$. A corresponding palaeomagnetic pole resides at 35.9°S and 217.9°E. This direction agrees with earlier studies of similar age rocks from the RIC.

(4) Considering an emplacement age of 932 ± 7 Ma, and a cooling rate of 15 °C myr^{-1} for the region, the age of remanence acquisition for the BKS is 916 Ma.

(5) The palaeolatitude of southern Norway would be 59.4°S at this time, indicating the position of Baltica is in high southern latitudes at this time.

(6) There are no geological, petrologic and geochemical or palaeomagnetic data to suggest that these stable palaeomagnetic directions are overprinted, or suffered later alteration since initial cooling of the igneous complex.

(7) In combination, with other published pole positions for Fennoscandia in the late Mesoproterozoic to early Neoproterozoic, it appears that the position of southwest Baltica was at a high latitude position post Grenville–Sveconorwegian orogenic activity, and remained in that position for tens of millions of years.

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